

ECE 443/518 – Computer Cyber Security

Lecture 10 Diffie-Hellman and Man-in-the-Middle Attack

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Diffie-Hellman Key Exchange

Man-in-the-Middle Attack

Reading Assignment

- ▶ This lecture: UC 8.1,8.5,13.3.1
- ▶ Next lecture: UC 10.1 – 10.3

Midterm Exam

- ▶ Lecture 1 ~ Lecture 14, see Homework 1 and 2 for sample.
 - ▶ Points may be deducted if key steps are missing.
- ▶ Students registered for main campus section: Wed. 10/14, 11:25 AM – 12:40 PM, in class.
 - ▶ A physical calculator is allowed. Laptop or any other electronic device or calculator apps running on them are not allowed.
 - ▶ Closed book/notes. A letter-size page of cheat sheet is allowed.
- ▶ Online students may take the exam as above, or
 - ▶ Reply the email on 9/10 from Charles Scott (scott@iit.edu) with the title “ECE 443/518 Exams for Online Section Students” to make arrangement, and cc me.
 - ▶ No make-up exam will be offered if you fail to do so.
- ▶ ADA Accommodations: contact Center for Disability Resource (disabilities@iit.edu)
- ▶ Emergency/extraordinary reasons for make-up midterm exams are accepted only with documented proof like doctor's notes.

Diffie-Hellman Key Exchange

Man-in-the-Middle Attack

Perfect Forward Secrecy (PFS)

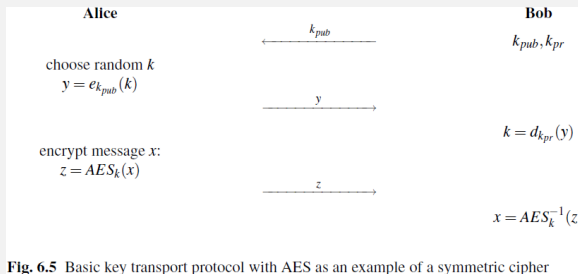


Fig. 6.5 Basic key transport protocol with AES as an example of a symmetric cipher

(Paar and Pelzl)

- ▶ Oscar, realizing there is no hope to factor every n for RSA, decided to build a machine to factor a few n 's he/she might be interested of.
 - ▶ Oscar has recorded all communications encrypted using the simple hybrid protocol.
- ▶ How could Bob protect k 's if k_{pr} is compromised?
 - ▶ Will it help if Alice/Bob generates another RSA key-pair randomly per communication?

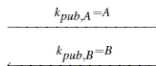
History of Diffie-Hellman Key Exchange

- ▶ 1976: created by Whitfield Diffie and Martin Hellman
 - ▶ First public-key algorithm in open literature.
- ▶ 1977: patent granted in US
- ▶ 2005: variants are included in NSA Suite B Cryptography
 - ▶ Together with AES and SHA-2

Diffie–Hellman Key Exchange

Alice
 choose $a = k_{pr,A} \in \{2, \dots, p-2\}$
 compute $A = k_{pub,A} \equiv \alpha^a \pmod{p}$

Bob
 choose $b = k_{pr,B} \in \{2, \dots, p-2\}$
 compute $B = k_{pub,B} \equiv \alpha^b \pmod{p}$



$$k_{AB} = k_{pub,B}^{k_{pr,A}} \equiv B^a \pmod{p}$$

$$k_{AB} = k_{pub,A}^{k_{pr,B}} \equiv A^b \pmod{p}$$

(page 207, Paar and Pelzl)

- ▶ DHKE setup
 - ▶ A large prime p and an integer α chosen from $2, 3, \dots, p-2$.
 - ▶ Usually chosen/published by a well-known entity and used by a large group of people.
- ▶ Key exchange: upon completion, a shared secret k_{AB} is established between Alice and Bob.
 - ▶ Assume one of the public key is sent over an authentic channel.
- ▶ Time complexity: $O(N^3)$.

Example

- ▶ $p = 29, \alpha = 2$.
- ▶ Alice: $k_{pr,A} = 5$
 - ▶ $k_{pub,A} = 2^5 \bmod 29 = 3$
- ▶ Bob: $k_{pr,B} = 12$
 - ▶ $k_{pub,B} = 2^{12} \bmod 29 = 7$
- ▶ Alice: $k_{AB} = (k_{pub,B})^{k_{pr,A}} \bmod p = 7^5 \bmod 29 = 16$
Bob: $k_{AB} = (k_{pub,A})^{k_{pr,B}} \bmod p = 3^{12} \bmod 29 = 16$

The Discrete Logarithm Problem

Given a prime number p , an integer $\alpha \in \{2, 3, \dots, p-2\}$, and an integer B , solve for an integer b ,

$$\alpha^b \equiv B \pmod{p}.$$

- ▶ A passive adversary may obtain $k_{pr,B}$ and then k_{AB} .
- ▶ Brute-force: compute $\alpha^k \bmod p$ for $k = 1, 2, \dots, p-1$
 - ▶ Time complexity: $O(2^N N^2)$.
- ▶ Better algorithm exists, but still of exponential time.
- ▶ The Diffie-Hellman problem: compute $\alpha^{ab} \bmod p$ given $\alpha^a \bmod p$ and $\alpha^b \bmod p$ with α and p known.
 - ▶ It is unknown if this could be done without solving discrete logarithm first.
 - ▶ DHKE is believed to be secure for large enough N .
- ▶ Will Alice be able to learn Bob's private key?

The Elgamal Encryption Scheme

- ▶ An extension of DHKE for encryption.
- ▶ After successfully completing DHKE, Alice sends $y = k_{AB}x \bmod p$ to Bob.
 - ▶ Plaintext $x \in \{1, 2, \dots, p-1\}$
 - ▶ Ciphertext $y \in \{1, 2, \dots, p-1\}$
- ▶ Bob decrypts y by solving $k_{AB}x \equiv y \pmod{p}$ for x via EEA or other algorithms.
- ▶ Ephemeral keys: k_{AB} should be used only once.
 - ▶ A passive adversary who learned a pair of x and y could solve $k_{AB}x \equiv y \pmod{p}$ for k_{AB} and decrypts all other ciphertext with the same k_{AB} .

Elgamal Encryption Protocol

Elgamal Encryption Protocol

Alice

choose $i \in \{2, \dots, p-2\}$
compute ephemeral key
 $k_E \equiv \alpha^i \bmod p$
compute masking key
 $k_M \equiv \beta^i \bmod p$
encrypt message $x \in \mathbb{Z}_p^*$
 $y \equiv x \cdot k_M \bmod p$

$\xleftarrow{k_{pub} = (p, \alpha, \beta)}$

$\xrightarrow{(k_E, y)}$

Bob

choose large prime p
choose primitive element $\alpha \in \mathbb{Z}_p^*$
or in a subgroup of $\mathbb{Z}_p^*$
choose $k_{pr} = d \in \{2, \dots, p-2\}$
compute $k_{pub} = \beta = \alpha^d \bmod p$

compute masking key
 $k_M \equiv k_E^d \bmod p$
decrypt $x \equiv y \cdot k_M^{-1} \bmod p$

- ▶ Rearrange messages: Elgamal looks very similar to RSA (page 228, Paar and Pelzl)
- ▶ Assume Bob's public key is sent over an authentic channel.
 - ▶ Alice's message could be sent over an insecure channel.
 - ▶ Need padding for similar reasons as RSA.

DHKE vs. RSA

- ▶ Good alternatives of each other.
 - ▶ Their security depends on different problems that we don't know how to solve efficiently yet.
 - ▶ While DHKE was originally designed for key exchange, its variants can match with the functionalities provided by RSA.
- ▶ Both DHKE and RSA may need an authentic channel for communicating a public key.
- ▶ In practice, both DHKE and RSA use keys a few thousand bits long to be secure.
- ▶ DHKE can be generalized to other mathematical structures.
 - ▶ E.g. elliptic-curve cryptography (ECC), which requires much less bits to achieve same level of security as DHKE, and is widely adopted currently.

Outline

Diffie-Hellman Key Exchange

Man-in-the-Middle Attack

The Authentic Channel

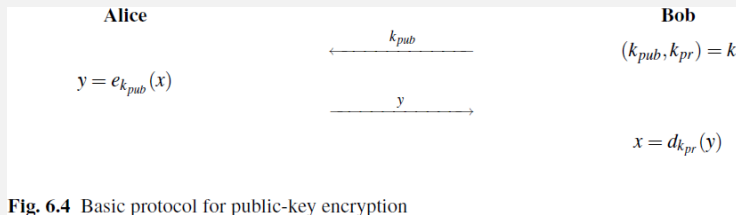


Fig. 6.4 Basic protocol for public-key encryption

(Paar and Pelzl)

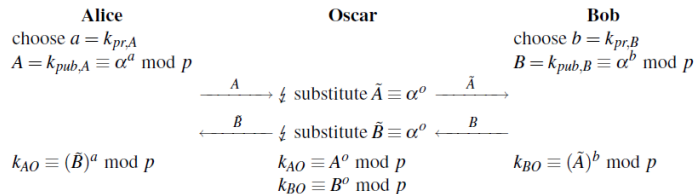
- ▶ We do see how RSA and DHKE (Elgamal) both use the above protocol for public key encryption.
- ▶ Both require Alice to receive Bob's public key on an authentic channel.
 - ▶ What if not?

Man-in-the-Middle Attack

- ▶ Assume Bob's public key $k_{pub,B}$ is sent through an insecure channel.
- ▶ Oscar the active adversary replaces $k_{pub,B}$ that Alice receives.
 - ▶ With Oscar's public key $k_{pub,O}$.
- ▶ Alice receives $k_{pub,O}$ and encrypts x as $y = e_{k_{pub,O}}(x)$.
- ▶ Oscar replaces y that Bob receives.
 - ▶ With $y' = e_{k_{pub,B}}(x)$.
 - ▶ Note that Oscar simply decrypts y to obtain x since y is encrypted with $k_{pub,O}$: $x = d_{k_{pr,O}}(y)$.
- ▶ Man-in-the-Middle: Oscar sits between Alice and Bob, and replaces all messages on either direction.
 - ▶ Neither Alice and Bob will be able to detect it!

Man-in-the-Middle and DHKE

Man-in-the-Middle Attack Against the DHKE



(Paar and Pelzl)

- ▶ This attack also applies to the original DHKE assuming both Alice and Bob's public keys are not sent via an authentic channel.
 - ▶ Oscar then have two secret keys, one with Alice and one with Bob, that can be used for any following communications.
- ▶ What if, as originally assumed, one of Alice and Bob's public keys is sent via an authentic channel?
- ▶ Does man-in-the-middle attack apply to symmetric ciphers?

Identity

- ▶ The problem of man-in-the-middle attack is with identity.
 - ▶ Alice sees Oscar as Bob.
 - ▶ Bob sees Oscar as Alice.
- ▶ The authentic channel authenticates that a public key belongs to a particular identity.
 - ▶ To create an authentic channel requires to establish identity – who is Bob?
- ▶ Can we establish identity without the authentic channel?
 - ▶ Yes if the public key is the identity, but how?
 - ▶ Indeed, in a successful man-in-the-middle attack, communications between Alice and Oscar is secure against any third party including Bob, and communications between Oscar and Bob is secure against any third party including Alice.

Summary

▶ DHKE

- ▶ Setup: prime p , $\alpha \in \{2, 3, \dots, p-2\}$.
- ▶ Alice: $k_{pr,A}$, publish $k_{pub,A} = \alpha^{k_{pr,A}} \bmod p$
- ▶ Bob: $k_{pr,B}$, publish $k_{pub,B} = \alpha^{k_{pr,B}} \bmod p$
- ▶ Alice and Bob: $k_{AB} \equiv (k_{pub,B})^{k_{pr,A}} \equiv (k_{pub,A})^{k_{pr,B}} \pmod{p}$
- ▶ Assumption: Oscar cannot solve $\alpha^b \equiv B \pmod{p}$ for b in polynomial time.

▶ Man-in-the-Middle attack